Evaluating Multimodal Large Language Models for Geospatial Object Identification and Enumeration in Overhead Imagery

Tim Klawa[1][0009-0009-8578-8688] and Frank Tanner[2][0009-0003-6814-2291]

[1] Groq, Public Sector, Field CTO

[2] Innodata, VP Robotics and Computer Vision

Abstract. This work explores the use of multimodal large language models (MLLMs) for geospatial object detection and enumeration in satellite imagery. Using the publicly available RarePlanes [1] dataset with pre-chipped satellite imagery, we evaluate nine frontier MLLMs on their ability to detect and classify aircraft in overhead imagery. Each model is provided with ontology class definitions and structured output requirements to identify and count aircraft across seven categories. We assess model performance on 300 representative image samples, measuring both object enumeration and classification accuracy. Our methodology examines model sensitivity to object scale, density, and visual context, and probes the extent to which MLLMs can support dynamic, zero-shot search without external embeddings or traditional computer vision models. By isolating the MLLM as a reasoning engine over visual data, this study informs the design of agentic pipelines for automated broad-area search, where adaptability, task generalization, and structured spatial reasoning are essential for defense and intelligence applications.

Keywords: Multimodal Large Language Models, Object Detection, Satellite Imagery, Aircraft Classification, Zero-Shot Learning, Geospatial Intelligence, Remote Sensing

1 INTRODUCTION

The rapid advancement of multimodal large language models (MLLMs) has opened new possibilities for automated geospatial intelligence analysis. Traditional computer vision approaches to object detection in satellite imagery rely heavily on supervised learning with annotated datasets, specialized architectures, and domain-specific feature extraction. While these methods have achieved remarkable success, they often require substantial training data, careful tuning for specific object classes, and they struggle to generalize to novel scenarios or rapidly changing requirements.

Recent developments in vision-language models, particularly GPT-4o, Claude Opus, Gemini, and Llama variants, demonstrate impressive zero-shot reasoning capabilities across diverse visual tasks. These models can interpret natural language instructions, understand spatial relationships, and identify objects without task-specific training.

This capability presents an intriguing question for the geospatial intelligence community: can MLLMs serve as standalone agents for broad-area search in overhead imagery, performing both detection and classification tasks with minimal adaptation?

This work presents a systematic evaluation of frontier MLLMs on aircraft detection and classification in satellite imagery using the RarePlanes [1] dataset. We benchmark nine models (GPT-4o, GPT-4o Turbo, GPT-4o Mini, Claude Opus 4.1, Claude Sonnet 4, Gemini 2.5 Flash, Llama 4 Maverick 17B, Grok-4 Fast Reasoning, and Grok-4 Fast non-Reasoning) across 300 image samples, measuring their ability to accurately count aircraft and classify them according to a defined taxonomy.

Our experimental framework isolates the MLLM as the sole reasoning component, avoiding external computer vision models or embedding systems. This approach allows us to assess the intrinsic capabilities of these models for geospatial analysis and identifies both their strengths and limitations in operationally relevant scenarios.

2 BACKGROUND AND RELATED WORK

2.1 Object Detection in Overhead Imagery

Object detection in satellite and aerial imagery presents unique challenges compared to terrestrial image understanding. These challenges include small object sizes relative to image resolution, high aspect ratios and arbitrary orientations, dense object arrangements in complex scenes, variable lighting conditions and atmospheric effects, and limited training data for rare or novel object types [8]. Traditional approaches have employed convolutional neural networks [2,3], YOLO architectures [4,5], and more recently, vision transformers [6,7] to address these challenges. However, these methods typically require extensive labeled training data and struggle to adapt to new object classes or changing mission requirements without retraining.

2.2 The RarePlanes Dataset

Released by In-Q-Tel in 2019, the RarePlanes dataset represents a significant resource for aircraft detection and classification research. The dataset contains 253 Maxar WorldView-3 images spanning 112 locations and 2,142 square kilometers, with 14,700 hand-annotated aircraft. Additionally, it includes over 50,000 synthetic images with approximately 630,000 synthetic aircraft annotations.

The dataset provides a hierarchical taxonomy for aircraft classification with multiple levels of granularity. For this study, we utilize the role-based classification level, which includes seven classes: Civil Large Transport, Civil Medium Transport, Civil Small Transport, Military Fighter, Military Bomber, Military Transport, and Military Trainer.



Fig. 1. Example images from RarePlanes [1]

2.3 Multimodal Large Language Models

The emergence of vision-language models has enabled new approaches to visual understanding tasks. Recent work has demonstrated that vision transformers excel at satellite object detection tasks, with detection transformer (DETR) architectures achieving state-of-the-art performance on remote sensing benchmarks including aircraft and ship detection [9,10]. Building on these advances, models such as GPT-4o, Claude, Gemini, and Llama variants combine large-scale language understanding with visual processing capabilities, allowing them to interpret images in the context of natural language instructions. These models have demonstrated remarkable zero-shot and few-shot learning capabilities across diverse domains, suggesting potential applications in specialized areas like geospatial intelligence where traditional supervised learning approaches may be limited by data availability or rapidly evolving requirements.

Traditional computer vision models trained on labeled datasets with bounding boxes or segmentation masks can infer the exact location of objects because their architectures preserve spatial hierarchies and are explicitly optimized to regress geometric coordinates through convolutional features, anchor priors, and Intersection over Union (IoU) based localization losses. This supervision teaches the model not only *what* is in the image but precisely *where* it is. In contrast, it is important to note that modern generative AI vision models, such as CLIP-style encoders, diffusion backbones, or MLLMs, are not trained on spatially explicit objectives and instead learn global, semantic embeddings that compress and mix spatial information through tokenization and attention. Without detection heads, coordinate regression losses, or locality-preserving operations, these models can identify objects conceptually but cannot output deterministic or pixel-accurate positions, making them fundamentally different from traditional computer vision systems designed for localization.

3 METHODOLOGY

3.1 Data Selection and Preparation

For this experiment, we utilized a subset of the RarePlanes dataset curated by the Roboflow community [11]. This subset contains 2,710 image chips from the full dataset, pre-processed with COCO format annotations. The use of pre-chipped images simplified the experimental workflow while maintaining consistency with standard object

detection evaluation protocols. From this subset, we selected 300 representative image samples for evaluation, ensuring coverage of varying aircraft densities, scene complexities, and lighting conditions. This sample size provides sufficient statistical power for comparative analysis while remaining computationally tractable for repeated evaluation across multiple models.

3.2 Model Selection

We evaluated nine frontier multimodal language models representing diverse architectural approaches and training paradigms: GPT-4o, GPT-4o Turbo and GPT-4o Mini (OpenAI), Claude Opus 4.1 and Claude Sonnet 4 (Anthropic), Gemini 2.5 Flash (Google), Llama 4 Maverick 17B (Meta), Grok-4 Fast Reasoning, and Grok-4 Fast non-Reasoning (xAI). We selected these models to provide a comprehensive comparison of leading AI labs against object identification and enumeration in overhead Imagery. These models also represent both open source and commercial variants with the inclusion of Llama 4 Maverick.

3.3 Experimental Framework

We developed a custom MLLM evaluation platform to systematically test and compare model performance. The platform enables dynamic configuration of system prompts, ontology definitions, and automated evaluation workflows against pre-labeled datasets, facilitating consistent multi-model benchmarking. The framework used for evaluation can be extended or applied to a wide range of models and is not tied to any particular AI lab or inference engine, with support for evaluating MLLMs hosted on AWS, Azure, Google, Groq, OpenAI, Anthropic, and xAI. The open-source repo for the experiment can be found on Github (https://github.com/timklawa/gen-AI-Image-Eval-Engine).

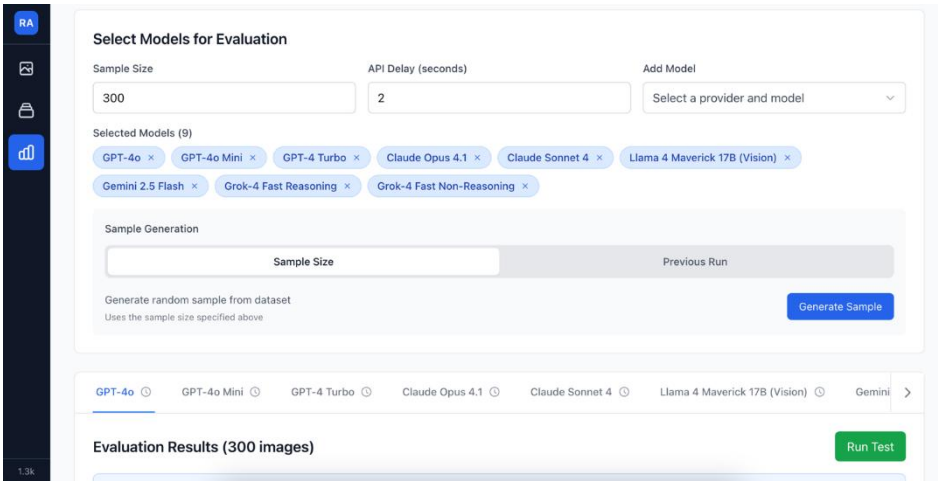


Fig. 2. Screenshot from evaluation platform

To provide consistency in comparing models, the same system prompt was used for each model evaluation. Each model was provided with a structured system prompt defining the task: *"You are an expert aircraft identification specialist. Analyze the provided satellite image and identify all aircraft present. For each aircraft, determine its class based on the provided ontology. Return your analysis in the specified JSON format with confidence scores for both count and individual class identifications."* The ontology class definitions were also provided to each model, specifying the seven aircraft categories from the RarePlanes taxonomy. Models were required to return structured JSON outputs containing total aircraft count, individual aircraft classifications, and confidence scores for both counting and classification.

3.4 Evaluation Metrics

We evaluated model performance across three primary dimensions: (1) Count Accuracy, the ability to correctly enumerate all aircraft in an image; (2) Classification Accuracy, the proportion of correctly classified aircraft according to the RarePlanes taxonomy; and (3) Processing Time, latency measurements to assess operational viability and consistency across repeated invocations.

Additionally, we analyzed model performance as a function of object density and examined failure modes through qualitative analysis of specific cases where models struggled or succeeded. There are several other metrics that could be pursued which we will describe in the Discussion section of this manuscript.

4 RESULTS

Our comprehensive evaluation of nine frontier MLLMs revealed significant insights into their capabilities for geospatial object detection and classification. Fig 3. provides the resulting performance across all evaluated models for count and class accuracy.

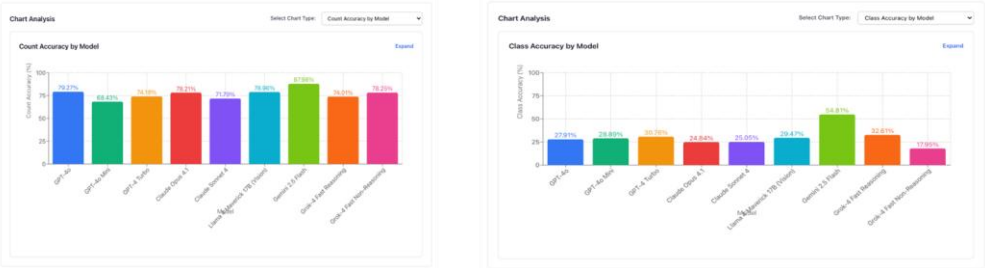


Fig. 3. Overall Model Performance Comparison. Left diagram shows count accuracy by model with right diagram showing classification accuracy by model

4.1 Object Count Accuracy

In general, our evaluation revealed some minor variation in counting accuracy across models as shown in Fig. 3 - Left. The Gemini 2.5 Flash model demonstrated the highest accuracy, achieving a peak count accuracy of 87.98%. A notable disparity was also observed within the GPT series; the GPT-4o model (79.27%) significantly outperformed its lightweight counterpart, GPT-4o Mini by more than 11 percentage points. Furthermore, the data presents an intriguing anomaly within the Grok models, as the Grok-4 Fast Non-Reasoning variant (78.25%) unexpectedly achieved a higher accuracy score than the Grok-4 Fast Reasoning model (74.01%).

4.2 Classification Accuracy

Classification performance proved more challenging for all models compared to object counting. It is important to note that MLLMs do not return bounding boxes like traditional CNNs; therefore, our classification analysis assumes the model is correctly identifying the pixels representing the plane itself, rather than another object, when assigning a class. Across the board, models struggled to accurately classify aircraft into the seven-category taxonomy (Fig. 3 – right). The Gemini 2.5 Flash model again achieved the highest accuracy by a significant margin, scoring 54.81%. Gemini 2.5 Flash's leading classification accuracy is likely dependent on its superior performance in the object count benchmark, given that successful class identification is inherently dependent on the accuracy of the preceding count. A significant reversal from the count accuracy results was observed in the Grok models; here, Grok-4 Fast Reasoning (32.61%) substantially outperformed the Grok-4 Fast Non-Reasoning variant (17.95%), highlighting the potential variance in reasoning as it relates to counting objects vs. classification tasks. Furthermore, the performance delta between GPT-4o (27.91%) and GPT-4o Mini

(28.89%) diminished almost entirely compared to the object count benchmark, with the 'Mini' model showing a slight advantage. Notably, the open-source Llama 4 Maverick 17B (Vision) model (29.47%) also proved competitive, surpassing the performance of both GPT-4o (27.91%) and Claude Sonnet 4 (25.05%) in this analysis.

On reviewing results from each model, we found that certain classes were easier for models to detect than others. Overall, distinguishing between classes that were size-related (Large vs. Medium vs. Civil Small Transport) proved challenging on a number of images, as did differentiating certain military classes. This is likely because the ontology lacked explicit size definitions and the image chips did not include a pixel-to-meter ratio (i.e. ground sample distance) to support size-based classification. Additionally, military classes were highly under-represented in the dataset. Our hypothesis is that MLLMs that struggled with military classifications likely had fewer relevant examples in their respective training datasets.

4.3 Object Density vs Accuracy



Fig. 4. number of objects per image vs model accuracy

As shown in Fig. 4, across both charts (Object Count vs. Count Accuracy and Object Count vs. Class Accuracy) the data reveals that there is no consistent relationship

between the number of objects present in an image and the corresponding model accuracy. Our initial hypothesis was that as object count increased, model performance, particularly for reasoning-intensive tasks like classification, would degrade due to higher visual and cognitive load. However, the results challenge this assumption: even at higher object densities (above 20 objects), several models maintained comparable count and class accuracy levels to those observed in simpler scenes. Interestingly, while accuracy variance appears wider in high-object scenarios, no single model demonstrates systematic degradation tied to object quantity. Instead, performance differences seem to stem more from model architecture and reasoning capability than sheer object complexity. This suggests that, for multi-object geospatial scenes such as those in the RarePlanes dataset, scene density alone is not a reliable predictor of MLLM performance degradation—a counterintuitive finding that warrants deeper investigation into factors such as geospatial understanding, relational reasoning, and compositional generalization. It is important to note that the RarePlanes dataset contains a higher proportion of low-density images compared to high-density ones, and that factors such as lighting conditions, background clutter, and overall scene complexity are not explicitly represented or controlled within the dataset's annotations.

4.4 Processing Time and Operational Considerations

The API Time per Image chart reveals substantial latency variability across models, with the Gemini 2.5 Flash model exhibiting particularly large and inconsistent spikes in processing time. This volatility highlights a key challenge for operational deployment of MLLMs in real-time or mission-critical contexts: ensuring predictability and consistency in inference time. The observed performance irregularities underscore the importance of continued research into deterministic compute paradigms, such as fixed-latency architectures or compiler-level scheduling optimizations, to achieve reproducible and time-bounded inference at scale.

The latency variations across the models tested may also stem from resource contention or dynamic load balancing within shared cloud environments, where compute and memory availability fluctuate based on concurrent demand. Such dependency on variable infrastructure reinforces the importance of dedicated or isolated model deployments particularly for mission-critical applications where consistent performance, deterministic timing, and reliability are essential for operational assurance and real-time decision-making.

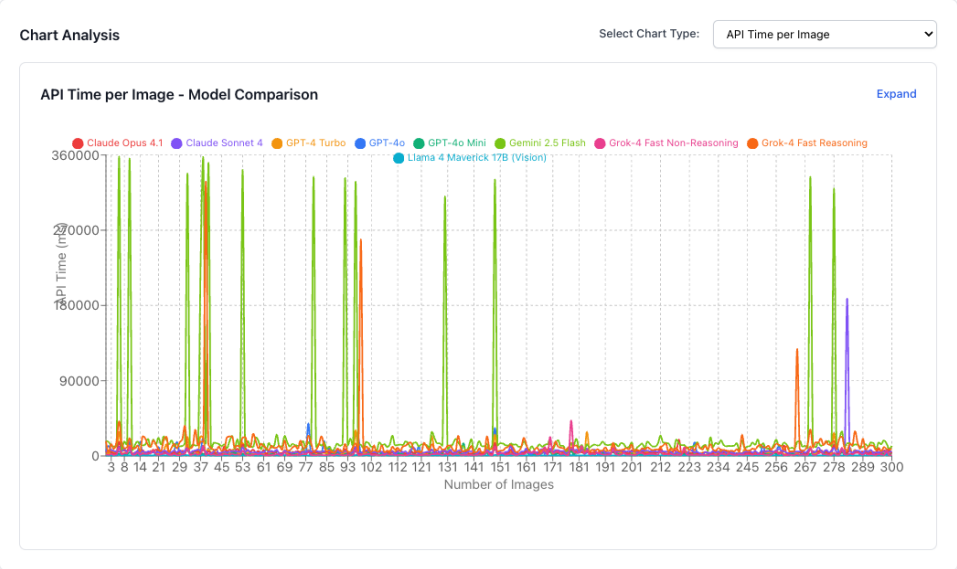


Fig. 5. Processing time by model

5 DISCUSSION

5.1 Key Findings

Our evaluation of nine frontier multimodal large-language models (MLLMs) for overhead imagery enumeration and classification reveals several important patterns. Overall, current-generation models exhibit meaningful competence at object counting, demonstrating that zero-shot reasoning alone can support broad-area search and change-detection tasks. However, fine-grained classification remains a significant challenge, particularly for visually similar military aircraft categories. This disparity suggests that enumeration leverages low-level saliency detection that these models already possess from general vision training, while classification depends on specialized spatial and geometric cues largely absent from their pre-training corpora.

Error analysis also indicates that the remaining mistakes were not purely "missed detections." Instead, they often involved boundary ambiguities (e.g., partial objects at chip edges) or category confusions such as distinguishing small civil transports from trainers. These outcomes imply that performance ceilings are now dictated less by coarse detection capacity and more by spatial reasoning and semantic disambiguation within the geospatial domain.

5.2 Factors Affecting Performance

Several qualitative tendencies emerged in our evaluation that suggest how scene characteristics influence model behavior. Larger and more visually distinct aircraft

were generally counted more accurately than smaller or partially obscured ones. In contrast, aircraft near chip boundaries or those partially hidden by shadows or structures were detected less consistently. These observations indicate that model sensitivity may depend on object scale and visibility, even though these factors were not explicitly varied in our design.

Scenes containing multiple closely spaced aircraft also produced more variability in counts, suggesting that high object density may increase the likelihood of undercounting or merged detections. Conversely, images with sparse or well-separated aircraft yielded more stable enumeration results.

Several models appeared to incorporate contextual information such as runways, taxiways, or nearby infrastructure when estimating class type. While this behavior sometimes improved classification consistency, it also led to occasional misclassifications when background cues conflicted with object appearance.

It is important to note that all imagery in this study was drawn from Maxar WorldView-3 collects with consistent ground-sample distance, illumination, and collection geometry, and with minimal atmospheric distortion. This uniformity simplified our analysis by reducing uncontrolled variables, but it also limits generalization. The models were evaluated under highly consistent, high-quality imaging conditions, so their performance under more variable sensors, resolutions, or environmental conditions remains to be tested.

Taken together, these observations suggest that MLLM performance is influenced by object visibility, density, and surrounding context. However, because our experiment did not explicitly manipulate these variables, future studies should quantify their effects through controlled variation of scale, scene complexity, and sensor characteristics.

5.3 Comparative Insights with Recent Work

Our observations align closely with recent literature on object counting and small-object detection in remote-sensing imagery. Guo et al. [12] demonstrate that scale variation and attention design remain primary constraints on accurate enumeration, even in conventional deep-learning models. Hua et al. [13] likewise emphasize that object size, density, and occlusion influence aerial-detection performance, consistent with the qualitative trends observed in our evaluation.

Related studies in applied remote sensing also echo these findings. Delplanque et al. [14] show that object-counting models applied to wildlife surveys experience similar degradation under occlusion and background clutter, reinforcing that these limitations are not unique to aircraft imagery but may be intrinsic to high-resolution overhead analysis. Collectively, these peer-reviewed works support our conclusion that generalist MLLMs exhibit useful counting ability but are not yet reliable for fine-grained, context-independent classification.

5.4 Viability for Broad-Area Search and Operational Deployment

From an operational perspective, MLLMs show tangible promise for broad-area screening tasks. Their ability to rapidly enumerate and highlight probable object clusters suggests strong potential as first-pass filters in human-in-the-loop workflows. In such hybrid systems, models perform initial triage by flagging candidate sites or estimating object counts while trained analysts validate results or feed detections into specialized vision pipelines. This division of labor leverages the models' zero-shot adaptability and reduces analyst workload without compromising quality.

Utilization of MLLMs in their current state for broad-area search remains premature without fine-tuning or additional prompt engineering, such as providing example images per class for geospatial context. Unlike traditional computer vision models that generate explicit spatial outputs (e.g., bounding boxes or segmentation masks), MLLMs infer object presence through language-based reasoning over visual embeddings, which yields coarse approximations rather than precise locations. Their confidence scores also tend to cluster toward high values, even when incorrect, due to the intrinsic calibration limitations of generative models. These factors introduce operational risk and reinforce the need for structured prompting, domain-specific calibration, and further research in enabling realistic confidence judgements from MLLMs to unlock better human machine teaming capabilities for complex edge cases within mission datasets.

5.5 Limitations

Our study's limitations should guide interpretation of the results.

1. Dataset Uniformity: Because all imagery originated from Maxar WorldView-3 collects with consistent Ground Sample Distances (GSDs) and minimal atmospheric effects, the dataset represents a best-case scenario. Generalization to imagery from other sensors, altitudes, or environmental conditions remains untested. Consequently, our accuracy metrics likely represent an *upper bound* on achievable performance under ideal imaging conditions.
2. Sample Size and Scope: The evaluation encompassed 300 images which is sufficient for comparative analysis but limited for statistical generalization across global geographies or sensor modalities.
3. Zero-Shot Protocol: We intentionally avoided iterative prompt optimization to measure intrinsic capabilities. More aggressive prompt engineering or few-shot conditioning [15], which provides models with a small number of example inputs and outputs to guide task behavior, could yield higher accuracies.
4. Computational Variability: Latency fluctuations among API invocations likely reflect external infrastructure effects rather than intrinsic model performance but remain relevant to real-time use cases.

Taken together, these limitations highlight areas where future studies can extend this work, particularly by increasing dataset diversity, evaluating generalization to other sensors, and experimenting with adaptive prompting strategies.

6 CONCLUSIONS

This study demonstrates that multimodal large language models already possess meaningful intrinsic capability for geospatial object detection and enumeration in overhead imagery, achieving reasonable counting accuracy without any task-specific training. Yet fine-grained classification accuracy remains limited, especially among visually similar and underrepresented military aircraft categories.

Because our evaluation used uniform, high-quality Maxar WorldView-3 imagery from the RarePlanes dataset, the reported accuracies likely represent performance under optimal conditions. Real-world operational imagery, characterized by varying sensor geometries, resolutions, and atmospheric interference, will almost certainly reduce accuracy unless models are explicitly adapted to those conditions.

The observed variability in processing time and classification confidence further indicates that while MLLMs are viable analyst-assist tools, they are not yet ready to function as fully autonomous geospatial agents. Nonetheless, their zero-shot adaptability, ability to process structured natural-language instructions, and capacity for contextual reasoning make them valuable building blocks in next-generation hybrid human-machine intelligence pipelines.

In summary, while current multimodal models demonstrate promising zero-shot enumeration capability, achieving operational-grade geospatial precision will require deliberate methodological improvements. Future work should prioritize stronger domain alignment and explicit spatial grounding, along with structured mechanisms that compensate for the absence of task-specific training data. One recommended direction is the development of agentic workflows in which models iteratively decompose tasks, cross-validate intermediate outputs, and request additional context rather than relying on single-step inference. Another is the creation of vectorized exemplars for each ontology class, including variations in size, pose, and sensor conditions, to better anchor model reasoning and reduce misclassification in out-of-distribution scenarios. Collectively, these research directions provide a promising path toward improving MLLM robustness and achieving more consistent performance on diverse and operationally realistic geospatial datasets.

References

1. Shermeyer, J., Hossler, T., Van Etten, A., Hogan, D., Lewis, R., Kim, D.: RarePlanes: Synthetic Data Takes Flight. In: *Proc. IEEE/CVF Winter Conf. on Applications of Computer Vision (WACV 2021)*, pp. 207–217 (2021). https://arxiv.org/abs/2006.02963
2. Girshick, R.: Fast R-CNN. In: *Proc. IEEE Int. Conf. on Computer Vision (ICCV 2015)*, pp. 1440–1448 (2015). DOI: 10.1109/ICCV.2015.169

3. Lin, T.Y., Dollár, P., Girshick, R., He, K., Hariharan, B., Belongie, S.: Feature Pyramid Networks for Object Detection. In: *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR 2017)*, pp. 936–944 (2017). DOI: 10.1109/CVPR.2017.106
4. Redmon, J., Divvala, S., Girshick, R., Farhadi, A.: You Only Look Once: Unified, Real-Time Object Detection. In: *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR 2016)*, pp. 779–788 (2016). DOI: 10.1109/CVPR.2016.91
5. Cheng, T., Song, L., Ge, Y., Liu, W., Wang, X., Shan, Y.: YOLO-World: Real-Time Open-Vocabulary Object Detection. In: *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR 2024)*, pp. 16901–16911 (2024). https://arxiv.org/abs/2401.17270
6. Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., Guo, B.: Swin Transformer: Hierarchical Vision Transformer Using Shifted Windows. In: *Proc. IEEE/CVF Int. Conf. on Computer Vision (ICCV 2021)*, pp. 10012–10022 (2021). https://arxiv.org/abs/2103.14030
7. Aleissaee, A.A., Kumar, A., Anwer, R.M., Khan, S., Cholakkal, H., Xia, G.-S., Khan, F.S.: Transformers in Remote Sensing: A Survey. *Remote Sensing* 15(7), 1860 (2023). DOI: 10.3390/rs15071860
8. Tanner, F., Colder, B., Pullen, C., Heagy, D., Eppolito, M., Carlan, V., Oertel, C., Sallee, P.: Overhead Imagery Research Data Set — An Annotated Data Library and Tools to Aid in the Development of Computer Vision Algorithms. In: *Proc. IEEE Applied Imagery Pattern Recognition Workshop (AIPR 2009)*, pp. 1–8 (2009). DOI: 10.1109/AIPR.2009.5466304
9. Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A., Zagoruyko, S.: End-to-End Object Detection with Transformers. In: Vedaldi, A., Bischof, H., Brox, T., Frahm, J.M. (eds.) *European Conf. on Computer Vision (ECCV 2020)*, LNCS 12346, pp. 213–229. Springer (2020). https://arxiv.org/abs/2005.12872
10. Zhang, C., Su, J., Ju, Y., Lam, K.M., Wang, Q.: Efficient Inductive Vision Transformer for Oriented Object Detection in Remote Sensing Imagery. *IEEE Trans. Geosci. Remote Sens.* 61, 5616320 (2023). DOI: 10.1109/TGRS.2023.3292418 https://arxiv.org/abs/2306.12184
11. RarePlanes Dataset. Roboflow Universe (2023). Available at: https://universe.roboflow.com/home-workspace/rareplanes-1qgok (accessed 2 Nov 2025).
12. Guo, X., Anisetti, M., Gao, M., Jeon, G.: Object Counting in Remote Sensing via Triple Attention and Scale-Aware Network. *Remote Sensing* 14(24), 6363 (2022)
13. Hua, W., Chen, Q.: A survey of small object detection based on deep learning in aerial images. *Artificial Intelligence Review* 58, Article 162 (2025).
14. Delplanque, A., Théau, J., Foucher, S., Serati, G., Durand, S., Lejeune, P.: Wildlife detection, counting and survey using satellite imagery: Are we there yet? *GIScience & Remote Sensing* 61(1), Article 2348863 (2024).
15. Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., et al.: Language Models are Few-Shot Learners. *Advances in Neural Information Processing Systems (NeurIPS 2020)*, 33, 1877–1901 (2020).